

Predicting Diabetes Readmissions in U.S Hospitals

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1) Introduction

Hospital readmissions remain a significant challenge for healthcare systems in the United States. Patients with chronic conditions, particularly diabetes, contribute heavily to these readmissions, which increases healthcare costs and reduces hospital capacity. As a result, identifying patients at high-risk of readmission has become an important objective in healthcare analytics.

Advances in data science and machine learning provide powerful tools for addressing hospital readmissions. By leveraging large clinical datasets, predictive models can analyze relationships between patient demographics, diagnoses, medications, and prior healthcare utilization to estimate readmission risk and identify patients who may benefit from early intervention. This project applies machine learning to predict readmissions among patients with diabetes using real healthcare data. Building on the model developed in Capstone I, the project aims to improve performance and interpretability through continued preprocessing and explainability methods such as SHAP analysis, while also developing an interactive Streamlit dashboard that allows users to explore patient variables and view predicted readmission risk.

Overall, the goal of this project is to develop an interpretable machine learning solution for predicting diabetes-related hospital readmissions. By combining preprocessing, optimization, explainable AI, and deployment, this project illustrates how data science techniques can contribute to improving healthcare outcomes by reducing hospital readmissions.

2) Background

Hospital readmissions are often used as indicators of healthcare quality and care coordination. Because diabetes is associated with complications such as cardiovascular disease, kidney disease, and infection, diabetic patients face a higher likelihood of hospitalization and readmission. As a result, healthcare providers increasingly rely on predictive analytics and machine learning models to identify patients at high-risk of readmission (Halac, Cocucci, & Camerlingo, 2025).

The dataset used in this project is the UCI Diabetes 130-US Hospitals dataset, containing over 100,000 hospital encounters from 130 hospitals between 1999 and 2008 (Strack et al., 2014). It includes patient demographics, laboratory results, medication usage, diagnostic codes, and healthcare utilization measures. The original target variable had three classes (no readmission, readmission within 30 days, and readmission after more than 30 days). For this study, the two readmission categories were combined into a single readmitted class, creating a binary classification problem with approximately 54% non-readmitted

and 46% readmitted cases. This transformation simplified the modeling task, though a multiclass approach is still being evaluated for potential performance improvements.

Machine learning models are widely used for predicting hospital readmissions because they capture complex patterns in healthcare data and often outperform traditional statistical methods (Rajkomar, Dean, & Kohane, 2019). Models such as Random Forest, XGBoost, LightGBM, and SVMs perform well on heterogeneous medical datasets, but interpretability remains essential in healthcare settings. Explainable AI techniques such as SHAP estimate each feature's contribution to a prediction, improving transparency in model decisions (Lundberg & Lee, 2017). In this project, SHAP analysis identified emergency visits and inpatient stays as key drivers of readmission risk.

3) Methodology

This project follows a structured machine learning workflow to predict hospital readmissions among patients with diabetes, including data preprocessing, model training, evaluation, and deployment stages commonly used in healthcare analytics (Stanford University School of Medicine, 2020). Building on the Capstone I model, the methodology extends the analysis through improved preprocessing, model comparison, explainable artificial intelligence (XAI), and initial deployment. Several algorithms were evaluated, including Random Forest, XGBoost, LightGBM, and Support Vector Machines, which are well suited for structured tabular data and nonlinear relationships.

After initial model training, steps such as threshold adjustments and feature engineering were applied to improve model performance and reliability. Because interpretability is essential in healthcare applications, Explainable Artificial Intelligence (XAI) techniques were incorporated into the methodology. Shapley Additive exPlanations (SHAP) were used to analyze global feature importance and patient-level predictions, providing insight into how the model identifies readmission risk. The project also explores deployment through a Streamlit-based prototype dashboard demonstrating how the model could generate readmission risk predictions. Although the deployment is still under development, it provides a foundation for integrating predictive models into healthcare decision-support tools.

Found below, figure 1 illustrates the overall workflow used in this project. The following subsections describe each stage of this workflow in greater detail.

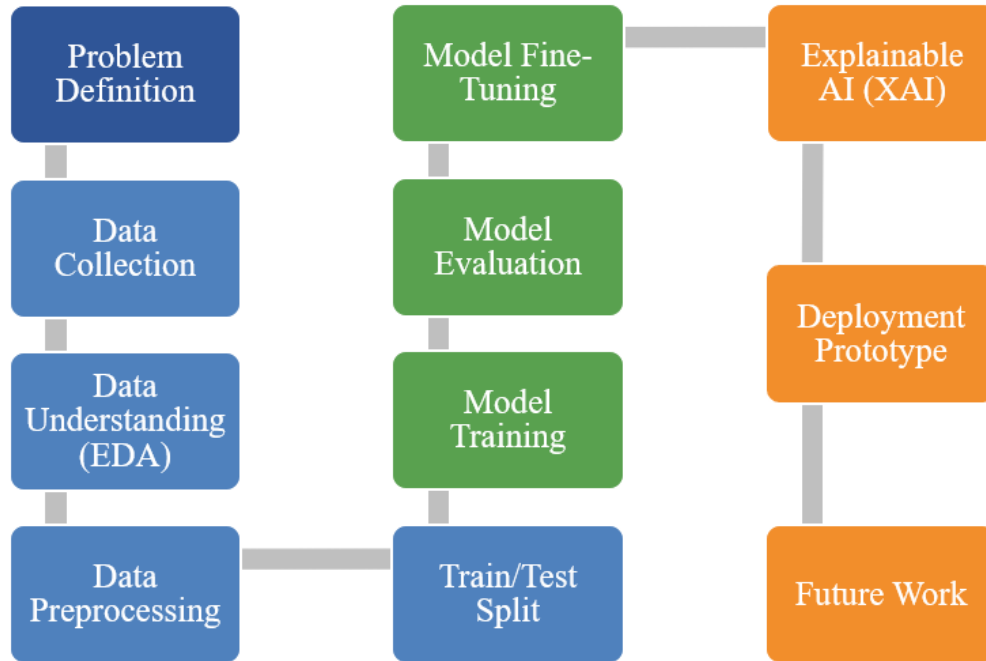


Figure 1. End-to-end machine learning pipeline used to develop, evaluate, and deploy the hospital readmission prediction model.

3.1) Data Understanding

The dataset used in this project is the Diabetes 130-US Hospitals dataset from the UCI Machine Learning Repository, containing over 100,000 hospital encounters from 130 U.S. hospitals between 1999 and 2008 (Strack et al., 2014). The dataset includes demographic variables, laboratory results, medication usage, and healthcare utilization measures such as emergency, inpatient, and outpatient visits.

The original target variable classified readmission into three categories: no readmission, readmission within 30 days, and readmission after more than 30 days. For this project, the two readmission categories were combined into a single readmitted class, creating a binary classification problem of readmitted vs. not readmitted. This transformation improved class balance to approximately 54% not readmitted and 46% readmitted and simplified the modeling task. Additional experiments are being conducted using the original three-class structure to evaluate whether retaining the original target structure can improve predictive performance.

Initial exploratory data analysis was performed to examine feature distributions, identify data quality issues, and analyze relationships between predictors and readmission outcomes.

3.2) Data Preprocessing

Data preprocessing prepared the dataset for machine learning modeling and addressed common healthcare data issues. Irrelevant identifiers were removed, missing values were examined, and categorical variables were standardized and encoded into numerical representations suitable for model training. The target variable was remapped from three categories (no readmission, <30 days, >30 days) into a binary outcome representing readmitted versus not readmitted, improving class balance. The dataset was then split into training and testing subsets to enable unbiased model evaluation.

3.3) Modeling

Multiple machine learning algorithms were evaluated to predict hospital readmission risk for diabetic patients, including Random Forest, XGBoost, LightGBM, and Support Vector Machines (SVM). These models were selected because they perform well on structured tabular data and can capture nonlinear relationships. Each model was trained on the preprocessed dataset and evaluated using an unseen test set with precision, recall, and F1-score, with particular emphasis on recall for the readmitted class. In healthcare applications, recall is especially important because failing to identify high-risk patients (false negatives) may prevent timely clinical intervention.

Initial model evaluation showed that ensemble tree-based methods performed most consistently, with LightGBM producing balanced results while maintaining strong recall for the readmitted class. On the test set, the model achieved a recall of 0.65, correctly identifying approximately 65% of readmitted patients. To further improve detection of high-risk patients, threshold tuning was applied, increasing recall to 0.74 while reducing precision to 0.61, indicating more false positives. In healthcare prediction tasks, this trade-off is often acceptable because identifying high-risk patients is prioritized over minimizing false alarms. Table 1 summarizes the performance of the machine learning models evaluated in this stage of the project.

Table 1. Performance comparison of machine learning models for predicting hospital readmissions.

Model	Precision (Readmitted)	Recall (Readmitted)	F1-Score (Readmitted)
Random Forest	0.63	0.63	0.63
XGBoost	0.63	0.64	0.64
SVM	0.54	0.60	0.57
LightGBM	0.64	0.65	0.64
LightGBM (Threshold tuned)	0.61	0.74	0.67

3.4) Fine-Tuning

After initial model evaluation, threshold tuning was applied to improve detection of high-risk patients. The classification threshold was adjusted from 0.50 to 0.45, increasing recall for the readmitted class to 0.74 while reducing precision to 0.61, reflecting an increase in false positives.

In the context of healthcare risk prediction, this trade-off is often acceptable because failing to identify a high-risk patient may lead to missed opportunities for early intervention (Rajkomar, Dean, & Kohane, 2019). Increasing recall allows the model to flag more patients who may benefit from follow-up care.

3.5) Explainable AI (XAI) Analysis

SHapley Additive exPlanations (SHAP) were used to analyze global and local model behavior. SHAP provides a tool for estimating the contribution of each feature to a prediction, allowing the model's prediction to be examined at both the population and patient levels.

Global model interpretability was evaluated using a SHAP Beeswarm plot (Figure 4, found in the “Key Results & Insights” section), which visualizes feature contributions across the dataset. As shown in Figure 4, features related to healthcare utilization, particularly number of emergency visits and number of inpatient visits, show the strongest influence on predictions. Additional features such as admission type, number of diagnoses, and insurance type also demonstrate influence on model predictions.

Local SHAP waterfall plots were generated for selected patient encounters to illustrate how individual features influence predictions. Higher emergency visit counts and inpatient history contributed positively to predicted readmission risk, while lower utilization values reduced predicted risk. Overall, the SHAP analysis indicates that the model relies on clinically meaningful utilization features rather than spurious correlations, supporting its potential use in healthcare settings.

3.6) Model Deployment

A preliminary Streamlit deployment was created to translate the modeling workflow into a web-based interface. The current prototype focuses on exploratory visualizations of the dataset, including distributions of key variables and sample data, while establishing the structure for future prediction features.

Future development will expand the dashboard to allow users to enter patient information and receive predicted readmission risk. The final version will also include SHAP visualizations to help explain model

predictions by showing how features such as emergency visits, inpatient history, and diagnostic information influence individual risk estimates. Developing this deployment framework early provides a foundation for integrating predictive models and explainability tools into an interactive dashboard.

[Link to preliminary Streamlit Dashboard.](#)

4) Challenges & Solutions

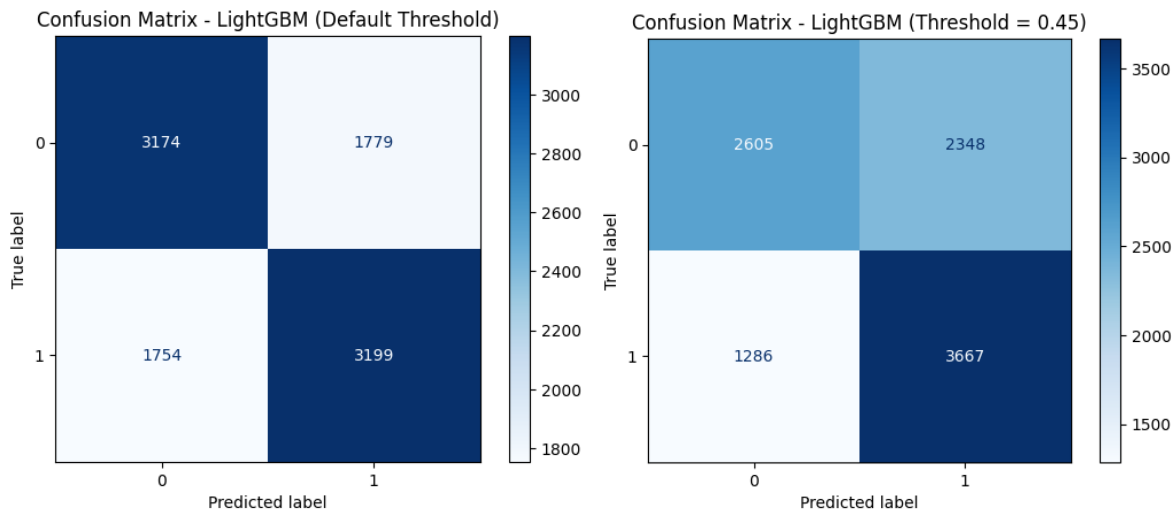
One challenge in this project was missing data, particularly for the *weight* variable, where more than 40% of values were missing. Because weight is an important clinical factor in Type 2 diabetes management, this level of missingness limits its usefulness in modeling. To reduce potential bias, the analysis prioritized features with more complete data, such as healthcare utilization and diagnostic variables. Another challenge involved the triple target variable (no readmission, <30 days, >30 days), which created class imbalance. To simplify the modeling task and improve stability, the two readmission categories were combined into a single readmitted class.

A final challenge involved balancing recall and precision when identifying high-risk patients. Through threshold tuning, recall for the readmitted class increased to 0.74, while precision decreased to 0.61. This trade off was considered acceptable because healthcare prediction models often prioritize sensitivity to ensure high-risk patients are identified for potential intervention (Rajkomar, Dean, & Kohane, 2019).

5) Key Results & Insights

The results show that LightGBM performed best for predicting hospital readmissions. After threshold tuning, the model achieved a recall of 0.74 for the readmitted class, improving its ability to detect patients who were later readmitted.

Figures 2 and 3 compare confusion matrices for the default threshold and the tuned threshold (0.45). Correctly identified readmissions increased from 3,199 to 3,667, an improvement of approximately 14.6%, while missed readmissions decreased from 1,754 to 1,286, a reduction of about 26.7%. This improvement came with an increase in false positives from 1,779 to 2,348 (~32% increase), reflecting the tradeoff between recall and precision.



Figures 2–3. Confusion matrices comparing LightGBM performance using the default classification threshold (0.50) and the tuned threshold (0.45).

Explainable AI analysis using SHAP showed that healthcare utilization features, particularly number of emergency visits and number of inpatient stays, were the strongest contributors to predicted readmission risk. These results suggest that the model relies on clinically meaningful patterns when identifying high-risk patients. The feature *number_inpatient* reflects prior hospital utilization and does not represent future information from the prediction period; therefore, it does not introduce data leakage.

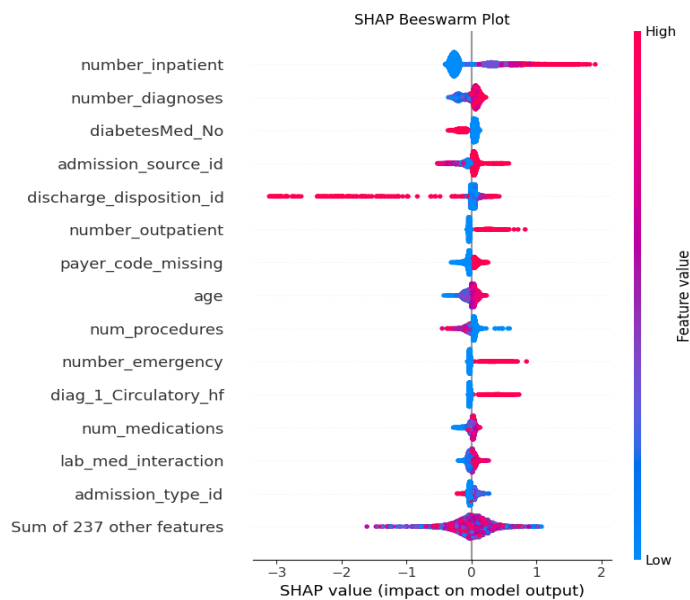


Figure 4. SHAP Beeswarm plot illustrating global feature importance for the hospital readmission prediction model.

6) Conclusion & Future Work

This project developed a machine learning framework for predicting hospital readmissions among diabetic patients using the UCI Diabetes 130-US Hospitals dataset. Several models were evaluated, with LightGBM producing the strongest performance. Threshold tuning improved recall for identifying high-risk patients, demonstrating the potential for machine learning to support hospital readmission risk assessment.

Future work will focus on testing models that retain the original three readmission classes (<30 days, >30 days, and no readmission) to evaluate whether separating these outcomes can improve predictive performance or provide more insights into readmission. Additional development will also focus on completing the interactive deployment dashboard and prioritize explainability to help users understand the factors driving model predictions.

Citations

- Halac, D., Cocucci, C., & Camerlingo, S. (2025). *Predictive machine learning model for 30-day hospital readmissions in a tertiary healthcare setting*. Bioinformatics Advances. <https://academic.oup.com/bioinformaticsadvances/issue/5/1>
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